Dr. Suman Das

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Department of Chemistry, GITAM
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Education:

- **Ph.D. Chemistry** 2016
  S. N. Bose National Centre for Basic Sciences (Jadavpur University), India
  Supervisor: Prof. Ranjit Biswas
  Thesis: Heterogeneity and its effects on simple chemical events in molten multi-component systems
- **M.Sc. Chemistry, 1st Class (C.P.I - 9.18)** 2011
  Indian Institute of Technology, Guwahati, India
- **B.Sc. Chemistry, 1st Class (67.6%)** 2009
  Ramakrishna Mission Vidyamandira, Belur (University of Calcutta)
- **Higher Secondary, 1st Class (89.7%)** 2006
  School - Singur Mahamaya High School
  Board - West Bengal Council of Higher Secondary Education
- **Secondary, 1st Class (91.6%)** 2004
  School - Singur Mahamaya High School
  Board - West Bengal Board of Secondary Education

Research Experience:

- **Postdoctoral Fellow**: University of Toronto, Toronto, Canada Dec. 2016 – March 2023
  Supervisor: Prof. Hue Sun Chan, Canada Research Chair (2001 – 2010)
  Research: Simulations of liquid-liquid phase separation in membraneless organelles
- **Doctoral Research**: S. N. Bose National Centre for Basic Sciences, India 2011 - 2016
  Supervisor: Prof. Ranjit Biswas
  Research- Heterogeneity in deep eutectic solvents (DESs) and ionic liquids (ILs)

- **M.Sc. Project**: Indian Institute of Technology, Guwahati, India 2010 - 2011

  Supervisor: Prof. Sandip Paul

  Research: Hydrogen bond properties and dynamics of liquid-vapour interface of water-DMSO mixtures

Research Interest:

- Sequence dependence of liquid-liquid phase separation in membraneless organelles
- Hydration dynamics around intrinsically disordered proteins
- Intrinsically disordered proteins bindings
- Quantum Computing
- Stability of proteins in non-aqueous solvents

Fellowships and Awards:

- Ramanujan Fellowship 2023, SERB
- Junior and Senior Research Fellow of Council for Scientific and Industrial Research (CSIR), New Delhi, India, December, 2010, **All India Rank – 63**
- Qualified Graduate Aptitude Test in Engineering (GATE) Fellowship, February, 2011, **All India Rank –704**
- Qualified Joint Admission Test for M.Sc. Admission (IIT - JAM), May 2009, **All India Rank – 184**

List of Publications:

1. Puvvada Roja Satvika, Ivy Das Sarkar, and **Suman Das***, Conformational Stability of Lysozyme in Choline Chloride-Polyol Aqueous Deep Eutectic Solvents (ADES), ***J. Chem. Phys. (just accepted).***
2. Priyanka Nath, Suman Choudhury, Huma Tabassum, Naveen Kumar, Sheetal Leva, Avijit Maity, Debanjan Bagchi, **Suman Das*, Anjan Chakraborty***, Metal-Ion-Modulated Stabilization of Intermediates in Diphenylalanine and Phenylalanine Self-Assembly: Roles of Metal-Ion-Specific Salt Bridges and Hydrophobic Interactions, **Biomacromolecules**. 2026, Q1, DOI: https://doi.org/10.1021/acs.biomac.6c01062, Journal Impact Factor-5.9.
3. Puvvada Roja Satvika, Ivy Das Sarkar, and **Suman Das***, Water-Modulated Structure and Dynamics in Aqueous Choline Chloride-Polyol Deep Eutectic Solvents: Insights from Molecular Dynamic Simulations, ***J. Chem. Phys.*** 2025, 163, 234506, Q1, DOI:10.1063/5.0299282, Journal Impact Factor-4.304.
4. Puvvada Roja Satvika, Preeti Sahu, **Suman Das***, and Anik Sen*, Understanding Transport and Structural Properties in Py-FSI–Alkali Metal Ion Mixtures: Role of Ion Identity and TIP4P-based Water Potential, **J. Chem. Eng. Data,** 2025, 70, 4979-4993, Q2, DOI:10.1021/acs.jced.5c00596, Journal Impact Factor-2.1.
5. Tamaghna Maitra, Param Jeet Singh, Atanu Bhattacharya, **Suman Das**, K.K. Gorai, D.V. Udupa, Vacuum ultraviolet spectra of gas phase organophosphorus solvent for nuclear fuel applications: Di-(2-ethyl-hexyl) phosphoric acid, **Chem. Phys**, 599, 2025, Journal Impact Factor 2.4.
6. Ivy Das Sarkar[#], Arnab Sil[#], Biswajit Guchhait[*], and **Suman Das**[*], Hydrogen-bond induced non-linear size dependence of lysozyme under the influence of aqueous glyceline, **J. Chem. Phys.** 2025, 162, 114506, Journal Impact Factor-4.304.
7. Arnab Sil, Sangeeta Sangeeta, Vishnu Poonia, **Suman Das*** and Biswajit Guchhait*, Molecular dynamics

insights into the dynamical behavior of structurally modified water in aqueous deep eutectic solvents (ADES), **J. Chem. Phys.** 161, 164501, 2024, Journal Impact Factor-4.304.

8. Yi-Hsuan Lin, Tae Hun Kim, **Suman Das**, Tanmoy Pal, Jonas Wessén, Atul Kaushik Rangadurai, Lewis E Kay, Julie D Forman-Kay and Hue Sun Chan*, Electrostatics of Salt-Dependent Reentrant Phase Behaviors Highlights Diverse Roles of ATP in Biomolecular Condensates, **eLife** 2025, 13, RP100284. Journal Impact Factor 6.4.
9. Tanmoy Pal#, Jonas Wessen#, **Suman Das#**, and Hue Sun Chan*, Differential Effects of Sequence-Local versus Nonlocal Charge Patterns on Phase Separation and Conformational Dimensions of Polyampholytes as Model Intrinsically Disordered Proteins (# equal first authorship), **J. Phys. Chem. Lett.** 15, 8248–8256, 2024, Journal Impact Factor-5.7.
10. Atanu Bhattacharya*, Param Jeet Singh*, and **Suman Das**, 1UV-VUV Absorption Spectra of Azido-Based Energetic Plasticizer Bis(1,3-diazido prop-2-yl)malonate in Gas Phase, **J. Chem. Phys.** 160, 014303, 2024, Journal Impact Factor-4.304.
11. Arnab Sil, Sangeeta, Renu Bhati, **Suman Das*,** Biswajit Guchhait*, Ion clustering, aggregation and diffusion in amide based deep eutectic solvents: A microstructural investigation using various NMR spectroscopic techniques and molecular dynamics simulation, **J. Mol. Liq.** 2023, 388, 122761, Journal Impact Factor-6.633.
12. Tamisra Pal[#] , **Suman Das[#]** , Dhrubajyoti Maji and Ranjit Biswas*, Validity of the Onsager-Glarum Relation in a Molecular Coulomb Fluid: Investigation via Temperature-dependent Molecular Dynamics Simulations of a Representative Ionic Liquid, [BMIM][PF6], (# equal first authorship) **New J. Chem.** 2023, 47, 14906-14920, Journal Impact Factor-3.925.
13. Jonas Wessén[#], **Suman Das,** Tanmoy Pal, and Hue Sun Chan***,** Analytical Formulation and Field-Theoretic Simulation of Sequence-Specific Phase Separation of Protein-Like Heteropolymers with Short- and Long-Spatial-Range Interactions, **J. Phys. Chem. B,** 2022, 126, 9222-9245, Journal Impact Factor-3.466.
14. Yi-Hsuan Lin[#], Jonas Wessén[#], Tanmoy Pal[#], **Suman Das**, and Hue Sun Chan*, Numerical Techniques for Applications of Analytical Theories to Sequence-Dependent Phase Separations of Intrinsically Disordered Proteins, invited Book Chapter in volume "Phase-Separated Biomolecular Condensates" in Nature Springer's **Methods in Molecular Biology** (**Springer-Nature**), (# equal first authorship) ISBN:978-1-0716-2662-7.
15. Kallol Mukherjee, **Suman Das,** Juriti Rajbanshi, Ejaj Tarif, Anjan Barman, and Ranjit Biswas*, Temperature-Dependent Dielectric Relaxation in Ionic Acetamide Deep Eutectics: Partial Viscosity Decoupling and Explanations from the Simulated Single-Particle Reorientation Dynamics and Hydrogen-Bond Fluctuations, **J. Phys. Chem. B**, 2021, 125, 12552-12567, Journal Impact Factor-3.466.
16. Jonas Wessén[#], Tanmoy Pal[#], **Suman Das[#]**, Yi-Hsuan Lin, and Hue Sun Chan*, A Simple Explicit-Solvent Model of Polyampholyte Phase Behaviors and its Ramifications for Dielectric Effects in Biomolecular Condensates, **J. Phys. Chem. B**, 2021, 125, 4337-4358, Journal Impact Factor-3.466, (# equal authorship, selected for cover image).
17. Tanmoy Pal[#], Jonas Wessen[#], **Suman Das** and Hue Sun Chan*, Subcompartmentalization of polyampholyte species in organelle-like condensates is promoted by charge pattern mismatch and strong excluded-volume interaction, **Phys. Rev. E**, 2021, 103, 042406 (#equal authorship), Journal Impact Factor-2.707.
18. Swarup Banerjee, Pradip Kr. Ghorai, Suman Das, Juriti Rajbanshi, and Ranjit Biswas*, Heterogeneous dynamics, correlated time and length scales in ionic deep eutectics: Anion and temperature dependence, **J. Chem. Phys.** 2020, 153 (23), 234502, Journal Impact Factor-4.304.
19. **Suman Das**, Yi-Hsuan Lin, Robert M. Vernon, Julie D. Forman-Kay and Hue Sun Chan*, Comparative Roles of Charge, π, and Hydrophobic Interactions in Sequence-Dependent Phase Separation of Intrinsically

Disordered Proteins. **Proc. Natl. Acad. Sci. U.S.A**, 2020, 117, 28795-28805, Journal Impact Factor-12.779.

20. Alan Amin[#], Yi-Hsuan Lin[#], **Suman Das** and Hue Sun Chan*, Analytical Theory for Sequence-Specific Binary Fuzzy Complexes of Charged Intrinsically Disordered Proteins, **J. Phys. Chem. B** 2020, 124, 6709–6720 (# equal authorship, selected for cover image), Journal Impact Factor-3.466.
21. **Suman Das**, Alan Amin, Yi-Hsuan Lin and Hue Sun Chan*, Coarse-grained residue-based models of disordered protein condensates: Utility and limitations of simple charge pattern parameters, **Phys. Chem. Chem. Phys.** 2018, 20, 28558-28574, Journal Impact Factor-3.676.
22. Kallol Mukherjee, **Suman Das**, Ejaj Tarif, Anjan Barman and Ranjit Biswas*, Dielectric relaxation in acetamide+urea deep eutectics and neat molten urea: Origin of time scales via temperature dependent measurements and computer simulations, **J. Chem. Phys.** 2018, 149, 124501, Journal Impact Factor-4.304.
23. **Suman Das**, Biswaroop Mukherjee* and Ranjit Biswas*, Orientational dynamics in a room temperature ionic liquid: Are angular jumps predominant? **J. Chem. Phys.** 2018, 148, 193839, Journal Impact Factor-4.304.
24. **Suman Das**, Adam Eisen, Yi-Hsuan Lin and Hue Sun Chan*, A lattice model of charge-pattern-dependent polyampholyte phase separation, **J. Phys. Chem. B** 2018, 122, 5418-5431, Journal Impact Factor-3.466.
25. **Suman Das**, Biswaroop Mukherjee* and Ranjit Biswas*, Microstructures and their lifetimes in acetamide/electrolyte deep eutectics: Anion dependence, **J. Chem. Sci.** 2017, 129, 939-951, Journal Impact Factor-2.15.
26. **Suman Das**, Ranjit Biswas* and Biswaroop Mukherjee*, Collective dynamic dipole moment and orientation fluctuations, cooperative hydrogen bond relaxations and their connections to dielectric relaxation in ionic acetamide deep eutectics: Microscopic insight from simulations, **J. Chem. Phys.** 2016, 145, 084504, Journal Impact Factor-4.304.
27. **Suman Das**, Ranjit Biswas* and Biswaroop Mukherjee*, Orientational jumps in (acetamide + electrolyte) deep eutectics: Anion dependence, **J. Phys. Chem. B** 2015, 119, 11157-11168, Journal Impact Factor-3.466.
28. Anuradha Das, **Suman Das** and Ranjit Biswas*, Density relaxation and particle motion characteristics in a non-ionic deep eutectic solvent (acetamide + urea): Time-resolved fluorescence measurements and all-atom molecular dynamics, **J. Chem. Phys.** 2015, 142, 034505, Journal Impact Factor-4.304.
29. **Suman Das**, Ranjit Biswas* and Biswaroop Mukherjee*, Reorientational jump dynamics and its connections to hydrogen bond relaxation in molten acetamide: An all-atom molecular dynamics simulation study, **J. Phys. Chem. B** 2015, 119, 274-283, Journal Impact Factor-3.466.
30. Biswajit Guchhait, **Suman Das**, Snehasis Daschakraborty and Ranjit Biswas*, Interaction and dynamics of (alkylamide + electrolyte) deep eutectics: Dependence on alkyl chain-length, temperature, and anion identity, **J. Chem. Phys.** 2014, 140, 104514, Journal Impact Factor-4.304.
31. Anuradha Das, **Suman Das** and Ranjit Biswas*, Fast fluctuations in deep eutectic melts: Multi-probe fluorescence measurements and all-atom molecular dynamics simulation study, **Chem. Phys. Lett.** 2013, 581, 47-51, Journal Impact Factor-2.719.

Poster Presentation:

1. **Suman Das** and Ranjit Biswas*, Multi-component deep eutectics: A simulation study, Theoretical Chemistry Conference, Indian Institute of Technology, Guwahati, India, 2012.

2. **Suman Das** and Ranjit Biswas*, Multi-component deep eutectics: A simulation study, Bose Fest, S. N. Bose National Centre for Basic Sciences, Kolkata, India, 2012.
3. **Suman Das** and Ranjit Biswas*, Multi-component deep eutectics: A simulation study, Bose Fest, S. N. Bose National Centre for Basic Sciences, Kolkata, India, 2013.
4. **Suman Das** and Ranjit Biswas*, Multi-component deep eutectics: A simulation study, Symposium on Theoretical and Computational Chemistry, Trichy, India, 2013.
5. **Suman Das**, Biswaroop Mukherjee* and Ranjit Biswas*, Reorientational jump and heterogeneity in amide deep eutectics, Dynamics of Complex Chemical and Biological Systems, Indian Institute of Technology, Kanpur, India, 2014.
6. **Suman Das** and Ranjit Biswas*, Dynamic heterogeneity and multi-point correlations in amide deep eutectics, Theoretical Chemistry Symposium, National Chemical Laboratory, Pune, India, 2014.
7. **Suman Das**, Biswaroop Mukherjee* and Ranjit Biswas*, Reorientational jump dynamics and its connections to hydrogen bond relaxations in molten acetamide: An all-atom molecular dynamics simulation study, International Symposium on Advances in Spectroscopy and Ultrafast Dynamics, Indian Association for the Cultivation of Science, Kolkata, India, 2014.
8. **Suman Das**, Biswaroop Mukherjee* and Ranjit Biswas*, Angular jumps in acetamide+electrolyte deep eutectics: An atomistic simulation study, Asian Academic Seminar and School, Indian Association for the Cultivation of Science, Kolkata, India, 2015.
9. **Suman Das** and Ranjit Biswas*, Dynamic heterogeneity and multi-point correlations in amide deep eutectics, Joint EMLG/JMLG Annual Meeting, University of Rostock, Rostock, Germany, 2015.
10. **Suman Das** and Ranjit Biswas*, Dynamic heterogeneity and multi-point correlations in amide deep eutectics, Ultrafast Science, S. N. Bose National Centre for Basic Sciences, Kolkata, India, 2015.
11. **Suman Das**, Biswaroop Mukherjee* and Ranjit Biswas*, Large amplitude angular jumps in (BMIM+PF_6) ionic liquid, Recent Advances in Molecular Spectroscopy, University of Hyderabad, Hyderabad, India, 2016.
12. **Suman Das**, Biswaroop Mukherjee* and Ranjit Biswas*, Large amplitude angular jumps in (BMIM+PF_6) ionic liquid, Bose Fest, S. N. Bose National Centre for Basic Sciences, Kolkata, India, 2016.
13. **Suman Das**, Adam Eisen, Yi-Hsuan Lin and Hue Sun Chan*, Monte Carlo simulations of phase separation of intrinsically disordered Proteins: A preliminary report, Protein Folding Consortium, 2017, University of California, Berkeley, USA, 2017.
14. **Suman Das**, Adam Eisen, Yi-Hsuan Lin and Hue Sun Chan*, Monte Carlo simulations of phase separation of intrinsically disordered Proteins, Departmental Retreat, University of Toronto, Toronto, Canada, 2017.
15. **Suman Das**, Alan Amin, Yi-Hsuan Lin and Hue Sun Chan*, Liquid-liquid phase separation in intrinsically disordered proteins: Simulations from lattice and off-lattice models, Chemical Biophysics Symposium, University of Toronto, Toronto, Canada, 2018.
16. **Suman Das**, Alan Amin, Yi-Hsuan Lin and Hue Sun Chan*, Liquid-liquid phase separation in intrinsically disordered proteins: Simulations from lattice and off-lattice models, Departmental Retreat, University of Toronto, Toronto, Canada, 2018.
17. **Suman Das**, Alan Amin, Yi-Hsuan Lin and Hue Sun Chan*, Liquid-liquid phase separation in intrinsically disordered proteins: Simulations from lattice and off-lattice models, Biophysical Society Meetings. Baltimore, Maryland, USA, 2019.
18. **Suman Das**, Alan Amin, Yi-Hsuan Lin and Hue Sun Chan*, Liquid-liquid phase separation in intrinsically disordered proteins: Simulations from lattice and off-lattice models, Canadian Biophysical Society Meetings. Toronto, 2019.
19. **Suman Das**, Alan Amin, Yi-Hsuan Lin and Hue Sun Chan*, Coarse-grained simulations of disordered protein condensates, University of Toronto, 2019, Toronto (**Best poster presentation**)

20. **Suman Das**, Yi-Hsuan Lin, Robert M. Vernon, Julie D Forman-Kay and Hue Sun Chan*, Comparative Roles of Charge, π, and Hydrophobic Interactions in Liquid-liquid Phase Separations of DEAD-box helicase Ddx4, Protein Folding Consortium, 2021.
21. **Suman Das**, Yi-Hsuan Lin, Robert M. Vernon, Julie D Forman-Kay and Hue Sun Chan*, Comparative Roles of Charge, π, and Hydrophobic Interactions in Liquid-liquid Phase Separations of DEAD-box helicase Ddx4, Canadian Biophysical Society Meeting, 2021.
22. **Suman Das**, Yi-Hsuan Lin, Robert M. Vernon, Julie D Forman-Kay and Hue Sun Chan*, Comparative Roles of Charge, π, and Hydrophobic Interactions in Liquid-liquid Phase Separations of DEAD-box helicase Ddx4, CSMB Meeting, 2021.

Oral Presentation:

1. **Suman Das**, Biswaroop Mukherjee* and Ranjit Biswas*, Cluster size and lifetime distribution in (amide+electrolyte) deep eutectics: Insights from molecular dynamics simulation study, Bose Fest, S. N. Bose National Centre for Basic Sciences, Kolkata, India, 2014 (**Best Oral Presentation**).
2. **Suman Das,** Biswaroop Mukherjee* and Ranjit Biswas*, Orientational jumps in (acetamide+electrolyte) deep eutectics: Anion dependence, Bose Fest, S. N. Bose National Centre for Basic Sciences, Kolkata, India, 2015 (**Best Oral Presentation**).
3. **Suman Das,** Alan N. Amin, Yi-Hsuan Lin and Hue Sun Chan*, Coarse-grained simulations of disordered protein condensates: Effect of interaction potentials and charge pattern parameters, Chemical Biophysics Symposium, 2019, Toronto.
4. **Suman Das,** Alan N. Amin, Yi-Hsuan Lin and Hue Sun Chan*, Coarse-grained simulations of disordered protein condensates, University of Toronto, 2019, Toronto.

Invited Posters:

1. SoPhyC 2024, IIT Bombay
2. SoPhyC 2025, IIT Patna

Invited Talks:

1. IMSc Chennai, 2024

Programming Languages: Fortran, C, C++, Python

Simulations Softwares: DL_POLY, DL_Field, Gromacs, Hoomd-Blue, Lammps, VMD, Packmol, Gaussian.

PhD Students:

- Puvvada Roja Satvika 2024-ongoing

Project Students:

- Alan Amin, Undergraduate student , University of Toronto, Canada 2017 - 2018
- Andrea Guljas, Rotation Master's Student, University of Toronto 2018
- Arnab Sil, PhD Student, Shiv Nadar University, India 2021-present
- Sangeeta, PhD Student, Shiv Nadar University, India 2021-present

- Abhishek, Master’s Student, IIT Kanpur, India 2023-2024
- Padma Sree, Master’s Student, GITAM, India 2023-2024
- Satvika, Master’s Student, GITAM, India 2023-2024
- Pavan Sai Kumar, GITAM, India 2024-2025